THE LCS NETWORK

A Plain-English Guide to Reticulum, LXMF and LXST

Liberty Communication Systems, Inc. · lcs.network

This document explains what the LCS Network is, what makes it work, and what it needs from you. It assumes you have never heard of **Reticulum** before. Everything here applies directly to the network model on the LCS site and to the three LCS applications: **Liberty Chat**, **LCS MeshChat** and **Liberty Deck**.

Contents

1. The Three Layers in One Minute

The LCS Network is built from three open protocols stacked on top of each other. Each one does a single job.

Layer	Name	What it does	Everyday comparison
Bottom	Reticulum (RNS)	Moves encrypted packets between devices over any medium — radio, Wi-Fi, cable, internet. Handles addressing, routing and encryption.	The road system
Middle	LXMF	Carries messages: text, images, files, attachments and recorded voice clips. Stores messages for people who are offline.	The postal service
Top	LXST	Carries live audio streams for real-time voice calls.	The telephone

- Reticulum is the foundation — nothing works without it.
- LXMF and LXST both ride on Reticulum, so they inherit its encryption and its routing automatically.
- An **LXMF client** is any app that speaks LXMF. Liberty Chat, LCS MeshChat and Liberty Deck are all LXMF clients, which is why they can all talk to each other — and to third-party clients like Sideband and Nomad Network.

2. Reticulum — The Network Underneath Everything

Reticulum is a networking stack designed for links that are slow, unreliable, and untrusted. It replaces the assumptions of TCP/IP with ones that survive on a radio channel. **Mark Qvist** is the man who has invented and spent years developing this powerful tool that revolutionizes communications.

What makes it different

- **No addresses or ports.** Reticulum uses a single concept called a **destination**. There is no DHCP, no subnetting, no port forwarding.
- **No central authority.** There is no registry, no account server, and no gatekeeper. Anyone can join by turning a device on.
- **Medium agnostic.** The same packets travel over LoRa, packet radio, Wi-Fi, Ethernet, USB serial, Bluetooth LE, or the public internet — mixed freely in one network.
- **Encrypted by default.** Encryption is not an option you enable. Unencrypted traffic cannot even be forwarded across multiple hops.
- **Extremely low bandwidth tolerance.** Reticulum is designed to stay functional on links as slow as **5 bits per second**.
- **Portable destinations.** Your address is not tied to a location. Move from home Wi-Fi to a LoRa radio on a mountain and you keep the same address.

Why that matters for LCS

- A message can start on a phone over Wi-Fi, cross a LoRa link, hop through a packet radio channel, and finish on a laptop across the internet — with no configuration in the middle.
- Every relay in that chain forwards the traffic without being able to read it.

3. Identities, Public Keys and Addresses

Identity

- An **Identity** is a public/private key pair generated on your own device the first time you run an app.
- Nothing is registered anywhere. No phone number, no email, no account.
- The **private key** stays on your device. It decrypts your incoming messages and signs proof that you received them.
- The **public key** is what you share with the network so others can encrypt messages that only you can open.
- Identities are portable files. Back one up, import it on a new phone, and your address moves with you. Liberty Chat can hold several identities and switch between them.

> Whoever holds your identity file controls your address. Back it up somewhere safe, and treat it like a house key.

Destination (your address)

- A **destination** is an address an app listens on. Your LXMF messaging address is a destination.
- It is displayed as **32 hexadecimal characters** — a 16-byte (128-bit) hash, for example

  `13425ec15b621c1d928589718000d814`
- The hash is derived from the app name, the destination type, and your public key — so two people can use the same app and still have unique, unforgeable addresses.
- Think of it as an email address that nobody issued to you, nobody can revoke, and nobody can impersonate.

4. How Peers Find Each Other — The Announce System

Reticulum has no directory server. Discovery happens through **announces**.

What an announce is

An announce is a small signed packet that a device broadcasts to say “I exist, here is how to reach me.” It contains:

- The destination hash (the address)
- The public key for that destination
- Optional app data — in messaging apps, this is your **display name**
- A random blob so every announce is unique
- An **Ed25519 signature** over all of it, proving the announce is authentic

How it spreads

- Any **transport node** that hears an announce records which neighbour it came from and how many hops it has travelled, then re-broadcasts it after a short random delay.
- Nodes never learn the whole route — each one only knows **the next hop** toward a destination. There is no map of the network anywhere.
- Announces stop propagating after **128 hops**.
- Announce traffic is capped at **2% of interface bandwidth** by default, and closer destinations are prioritized, so a fast network cannot flood a slow LoRa segment.
- If a device does not have a path, it can send a **path request** and the network will resolve one on demand.

What this means in practice

- **If you never announce, nobody can reach you.** You are invisible until you press Announce.
- When you announce, your name appears in everyone else's **Peers** or **Contacts** list automatically.
- Re-announce whenever you change how you are connected — unplugging an RNode, leaving home Wi-Fi, switching to the TCP server. That is how the network learns the new path to you.
- Announce sparingly. Announcing once an app starts and every few hours is plenty. Announce spam gets down-prioritized by the network, which makes you **harder** to reach, not easier.

5. Two Ways to Move Data — Packets vs. Links

Reticulum can deliver information two ways. Your apps choose automatically, but the difference explains a lot of the behavior you will see.

	Single Packet (opportunistic)	Link (established channel)
Best for	One short message, sent and forgotten	Conversations, voice calls, file transfers
Setup cost	None — sent immediately	3 packets, 297 bytes total
Encryption	A fresh key derived for every single packet	One shared key negotiated for the session
Forward secrecy	Only when key ratchets are enabled	Always
Delivery receipt	Optional signed proof	Built in
Large data	No — one packet only	Yes, via Resources: auto-split, compressed, verified, reassembled
Cost to hold open	n/a	About 0.45 bits per second
Anonymity	Initiator stays anonymous	Initiator stays anonymous unless they choose to identify

- A **link** is not a physical connection. It is an encrypted tunnel that can span many hops and stay open for hours.
- Both ends prove their identity cryptographically before any data flows, so you know you are talking to the right person and not an impostor.
- Because links are so cheap to keep alive, even a slow 1200 bps packet radio channel can hold roughly 100 concurrent links and still have most of its capacity free for data.

6. LXMF — Text, Files and Voice Messages

LXMF (Lightweight Extensible Message Format) is the messaging protocol. It is what makes your address an address and your message a message.

Key properties

- Total protocol overhead is only **111 bytes** per message — small enough to cross LoRa or packet radio comfortably.
- Every message is timestamped and **Ed25519-signed**, so the sender cannot be forged.
- Messages carry a title, a body, and an open-ended **fields** section used for attachments, images, voice clips, location and telemetry.
- Encrypted messages can even be printed as a QR code or a text link and carried on paper.

- All LXMF clients interoperate. Liberty Chat, LCS MeshChat, Liberty Deck, Sideband and Nomad Network all exchange messages freely.

Propagation Nodes — messages for offline users

- A **propagation node** is a store-and-forward mailbox for the network.
- If the person you are messaging is switched off, the message is held at a propagation node and delivered the moment they come back.
- Propagation nodes automatically peer with each other and synchronize, forming a distributed, encrypted message store — there is no single mailbox server to lose.
- On the LCS Network, the **Command Center PRO Gateway** and the LCS server both run this role. LCS MeshChat can also run one on your own PC.

> A propagation node stores messages it cannot read. The contents stay encrypted to the recipient's key for the entire time they sit in storage.

7. LXST — Real-Time Voice

LXST (Lightweight Extensible Signal Transport) is the live-audio protocol. Where LXMF is email, LXST is the telephone.

- Real-time voice calls between LCS apps, end-to-end encrypted with forward secrecy.
- Latency can be under 10 milliseconds on a fast link.
- Two codec families, chosen to match the link speed:
    - **Codec2** — 700 bps to 3200 bps. Intelligible speech over LoRa and packet radio.
    - **Opus** — roughly 4.5 kbps to 96 kbps. Natural, high-quality voice over Wi-Fi, Ethernet or the internet.
- Codecs can be switched **mid-call** without dropping the call. Liberty Chat probes the link before dialing and offers a switch if the current codec is too demanding — and the change is signaled to the other end, including MeshChat and Sideband peers.

Voice over LoRa — the practical limit

- Live voice needs a fast LoRa preset. In Liberty Chat, **Short Fast** (about 10.9 kbps) is the slowest preset that can carry a live call.
- **Half-Duplex** calls are recommended when making calls over LoRa radio and can be set prior to initiating the call.
- **Long Fast** (about 1.07 kbps) is the LCS recommendation for general messaging range, but it is too slow for a live call.
- If the link cannot support a live call, use **push-to-talk voice messages** instead — these are recorded clips sent as LXMF messages, and they work on any link.
- Liberty Chat PTT has two modes: **Low bandwidth** (Codec2 1200, about 150 bytes/sec — for LoRa) and **High bandwidth** (Opus, about 3 KB/sec — for Wi-Fi and TCP).

8. Node Types on the LCS Network

Reticulum itself only distinguishes two things: every device is an **instance**, and some instances are also **transport nodes**. The rest of the roles below are jobs a device performs on top of that.

Node type	Role	On the LCS Network
Client node	An endpoint. Runs an app, holds your identity, sends and receives. Does not relay for anyone.	LCS Client RNode plugged into a phone or PC by USB-C or Bluetooth; a phone on Wi-Fi; Liberty Deck.
Transport node	A relay. Forwards other people's encrypted traffic one hop closer. Set by enable_transport = Yes.	Solar Transport Node or IPRNode — solar panel and radio, hung in a tree or on a roof to extend LoRa range or a transport IPRNode in AP mode or connected to local Wi-Fi network.
Propagation node	A mailbox. Stores encrypted messages for users who are offline and delivers them on return.	Command Center PRO Gateway, the LCS server, or LCS MeshChat running on your own machine.
Command Center PRO Gateway	All of the above plus a protocol bridge — it routes traffic between LoRa, packet radio, Wi-Fi and TCP/IP, and hosts a local web app.	The flagship LCS unit. Connect to its Wi-Fi and open liberty.local in a browser to manage it or start messaging immediately.

More on the Command Center PRO Gateway

- **Bridges every interface.** Traffic arriving on LoRa can leave over packet radio, Wi-Fi or the internet, and vice versa. This is what stitches the separate segments of your diagram into one network.
- **Two Wi-Fi modes.** In **AP mode** it creates its own Wi-Fi network for phones and laptops in the field. In **client mode** it joins your home Wi-Fi and reaches the LCS server over the internet.
- **Runs MeshChat locally.** No download required — a browser and liberty.local is enough to start communicating.
- **Holds mail for the whole segment.** Anyone offline gets their messages when they return.
- **Packet radio software modem.** A digital modem plus a PTT interface drives an ordinary push-to-talk radio, so no separate hardware TNC is needed.

9. Interfaces — The On-Ramps

An **interface** is how a device running Reticulum physically gets onto the network. A single device can run several at once, and Reticulum will use whichever is best for each destination.

Interface	Medium	Typical use on the LCS Network	Notes
AutoInterface	Local Wi-Fi or Ethernet	Phones and PCs joining a Command Center Gateway, or several devices in one house	Zero configuration. Finds peers on the same LAN automatically. Needs UDP ports 29716 and 42671 open.
RNode (LoRa)	902–928 MHz ISM, license-free	Client radios and solar transport nodes. Roughly 341 bps to 21.88 kbps.	Connect by USB-C, Bluetooth LE, or over Wi-Fi as an IP RNode (tcp://iprnode.local).
KISS / packet radio	VHF 150–174 or UHF 450–470 MHz	Long-haul link between Command Center Gateways using push-to-talk radios	Uses the gateway's software modem. Encrypted business-band use requires an FCC license.
TCP Client	Internet or private IP	Apps reaching public.lcs.network or a local iprnode.local:4545	The way to talk to people worldwide.
TCP Server	Internet or private IP	The LCS server, and any gateway or PC that wants others to connect in	Many clients can connect to one server at once.

Interface	Medium	Typical use on the LCS Network	Notes
I2P	Anonymizing overlay network	Reachability without a public IP, and hiding endpoint IP addresses	Requires an I2P router (i2pd) on the machine. Gives a persistent, portable address.
Bluetooth LE	Short-range personal radio	Liberty Chat talking to a nearby RNode or another Android device	Enabled out of the box in Liberty Chat.
Serial / USB-C	Direct cable	Client RNode wired to a phone or computer	Simplest and most reliable client connection.

> A fresh Liberty Chat install comes up on AutoInterface and Bluetooth LE only. Nothing reaches the internet until you add a TCP server or attach an RNode — that is deliberate.

10. Transport Modes — How Traffic Is Steered

Every interface can be given a **mode** that tells Reticulum how to treat the segment behind it. You rarely need to change these on a client, but they are what make gateways behave correctly.

Mode	Behaviour	Where it is used
full	Default. All discovery, meshing and transport is active.	Ordinary client devices and most links.
gateway	Also resolves unknown paths on behalf of the devices behind it.	The interface a Command Center Gateway faces its clients with.
access_point	Stays quiet — no automatic announces, short-lived paths — until someone actually uses it.	A wide-area radio interface where users appear briefly and leave.
roaming	Marks an interface that physically moves relative to the rest of the network.	The LoRa side of a vehicle-mounted or mobile gateway.
boundary	Marks a link to a very different network segment.	A gateway's internet connection when the rest of its network is LoRa.
internal	The slow or private side that sits behind a boundary.	Local devices that should not receive the full firehose of internet announces.

- Setting **boundary** on the internet side and leaving the LoRa side in **full** keeps a large internet network from swamping a slow local radio segment with announces.
- A device only relays for others if **enable_transport = Yes**. Client apps leave this off, which is why a phone does not drain its battery routing strangers' traffic.

11. The Three LCS Apps

All three are LXMF clients. They speak the same protocol, use the same address format, and interoperate completely — the difference is the hardware they run on.

	Liberty Chat	LCS MeshChat	Liberty Deck
Runs on	Android phones and tablets	Windows, Mac, & Linux	LCS handheld device — screen and keyboard built in

	Liberty Chat	LCS MeshChat	Liberty Deck
Interface	Native Android app	Desktop app, or a browser at localhost:8000	On-device firmware, no phone or PC needed
Connects via	Bluetooth LE, Wi-Fi (AutoInterface), USB-C RNode, TCP server	Wi-Fi/Ethernet (AutoInterface), USB or IP RNode, TCP server, I2P	Built-in LoRa radio, plus Wi-Fi where supported
Text messaging	Yes	Yes	Yes
Files and images	Yes	Yes — receive limit raised to 10 MB	Limited by device storage
Push-to-talk voice	Yes — Codec2 or Opus	Yes	Limited
Live voice (LXST)	Yes, with mid-call codec switching	Yes	Basic
Propagation node	Uses them	Uses them and can host one	Uses them
Extras	Maps, offline map downloads, location sharing, QR identity sharing, multiple identities, Nomad Network browsing	Peer list, announce history, Nomad Network browsing, local message database	Standalone operation — nothing else to carry

- **Liberty Chat** ships with LCS radio defaults — slot 51 at 914.875 MHz, 22 dBm transmit power, Long Fast preset — and with the LCS servers already listed when you add a TCP connection.
- **LCS MeshChat** is the desktop and server client. It is the right choice for a machine that stays on, because it can host a propagation node for everyone else.
- **Liberty Deck** is a self-contained handheld. It carries its own identity, its own radio and its own screen, so it works with no phone and no computer attached.

12. Voice, Text and Data at a Glance

Feature	Protocol	Works over LoRa?	Works over packet radio?	Works over Wi-Fi / TCP?
Text messages	LXMF	Yes — the core use case	Yes	Yes
Delivery receipts	Reticulum proofs	Yes	Yes	Yes
Images and file attachments	LXMF fields	Slow but works — keep files small	Slow	Yes
Push-to-talk voice clips	LXMF audio field	Yes — use Codec2 low-bandwidth mode	Yes	Yes — Opus high-bandwidth mode
Live voice calls	LXST	Only on fast presets such as Short Fast	Marginal	Yes
Location sharing / telemetry	LXMF fields	Yes	Yes	Yes
Offline delivery	LXMF propagation nodes	Yes	Yes	Yes
Nomad Network pages and files	Reticulum links	Yes	Yes	Yes

13. Security Summary

What protects your traffic

- **End-to-end encryption on everything.** Only the destination holding the matching private key can decrypt a message. Relays, gateways and propagation nodes cannot.
- **Forward secrecy.** Links negotiate fresh ephemeral keys via X25519 Diffie-Hellman, so recording today's traffic and stealing a key tomorrow does not reveal it.
- **Authenticity.** Every announce and every message is Ed25519-signed. Addresses cannot be spoofed.
- **Initiator anonymity.** The party who opens a link stays anonymous unless they choose to identify — and that identification happens inside the encrypted tunnel, visible only to the far end.
- **No accounts, no metadata harvest.** There is no sign-up, no phone number, no contact-list upload, and no central log of who talked to whom.
- **Optional network isolation.** An interface can be given a **network name** and **passphrase**. Every packet then carries an Interface Access Code and anything without the right code is dropped at the door — a private network on shared airwaves.

The cryptography, specifically

- **X25519** for key exchange (ECDH on Curve25519)
- **Ed25519** for signatures and delivery proofs
- **AES-256 in CBC mode** with PKCS7 padding for payload encryption, inside an encrypted token
- **HMAC-SHA256** for integrity, **HKDF** for key derivation, **SHA-256** and **SHA-512** for hashing

> LXMF does not implement its own cryptography — it uses Reticulum's, for both link-delivered and single-packet messages. Reticulum's authoritative primitives specify AES-256 in CBC mode, so **AES-256 is the correct figure for LCS messaging, voice and data**.

What it does not protect

- **Your device.** If someone has your unlocked phone or your identity file, they have your address.
- **The fact that you transmitted.** Radio is radio. Encryption hides content, not the existence of a signal or its direction.
- **Legality.** Encrypted traffic on the business band needs an FCC license. The 902–928 MHz ISM band used by LoRa does not, but rules vary by country and remain the operator's responsibility.

14. Five Use-Case Scenarios

Scenario 1 — Two client RNodes with a solar transport node on the mountain

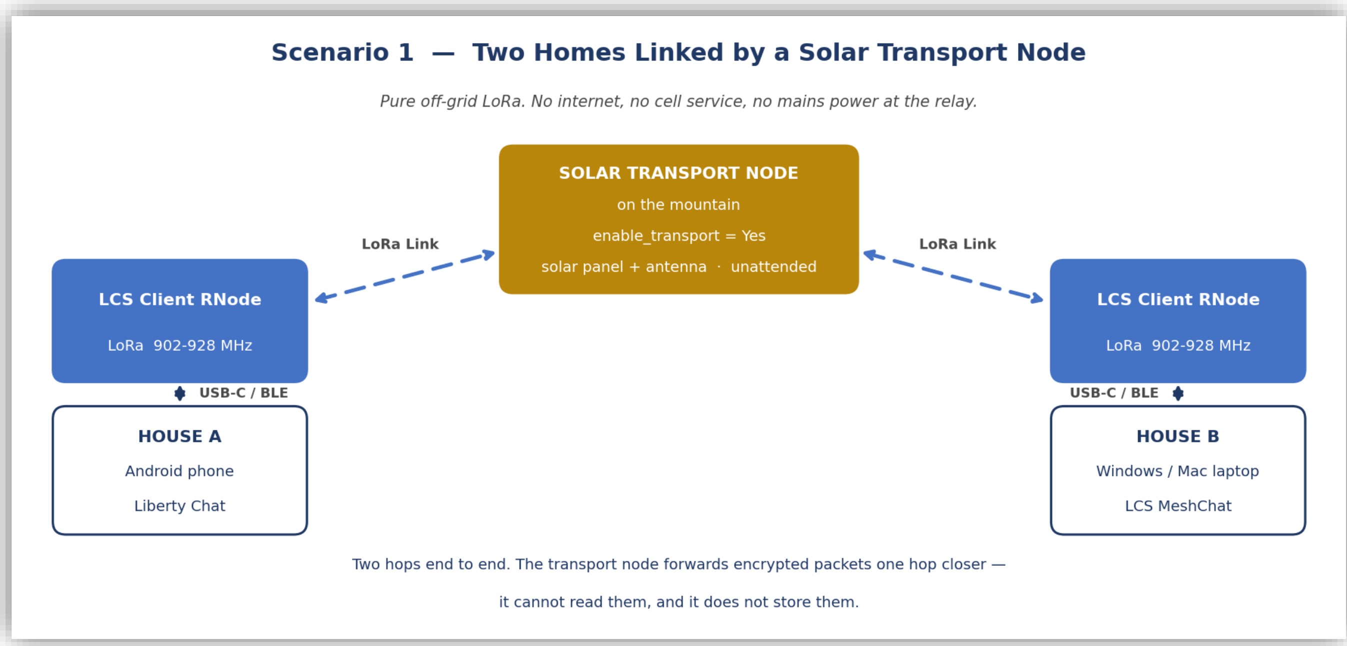

Figure 1 — Off-grid LoRa messaging across a relay.

What you need

- Two LCS Client RNodes, one per household, each on USB-C or Bluetooth to a phone or PC
- One LCS Transport/Repeater Node with solar panel and antenna, sited high with line of sight to both homes
- Liberty Chat on the phones, LCS MeshChat on the computers — same frequency and preset on every radio

How it works

- The mountain node has **enable_transport = Yes**, so it re-broadcasts announces and forwards packets. Nothing else in the chain needs to know it exists.
- Each user presses **Announce** once. The announce crosses the relay and both households appear in each other's peer list with a display name.
- Messages then route automatically: two hops, end to end encrypted, with signed delivery receipts.

Practical notes

- Use **Long Fast** for maximum range on text. Text messaging and PTT voice clips work well here.
- Live voice calls will not work at Long Fast speeds — switch to **Short Fast** and accept shorter range, or use half-duplex instead.
- The relay is unattended and has no mailbox. If a household is switched off, the sender's app retries when it sees a fresh announce.

Scenario 2 — Liberty Chat and LCS MeshChat connected through the internet

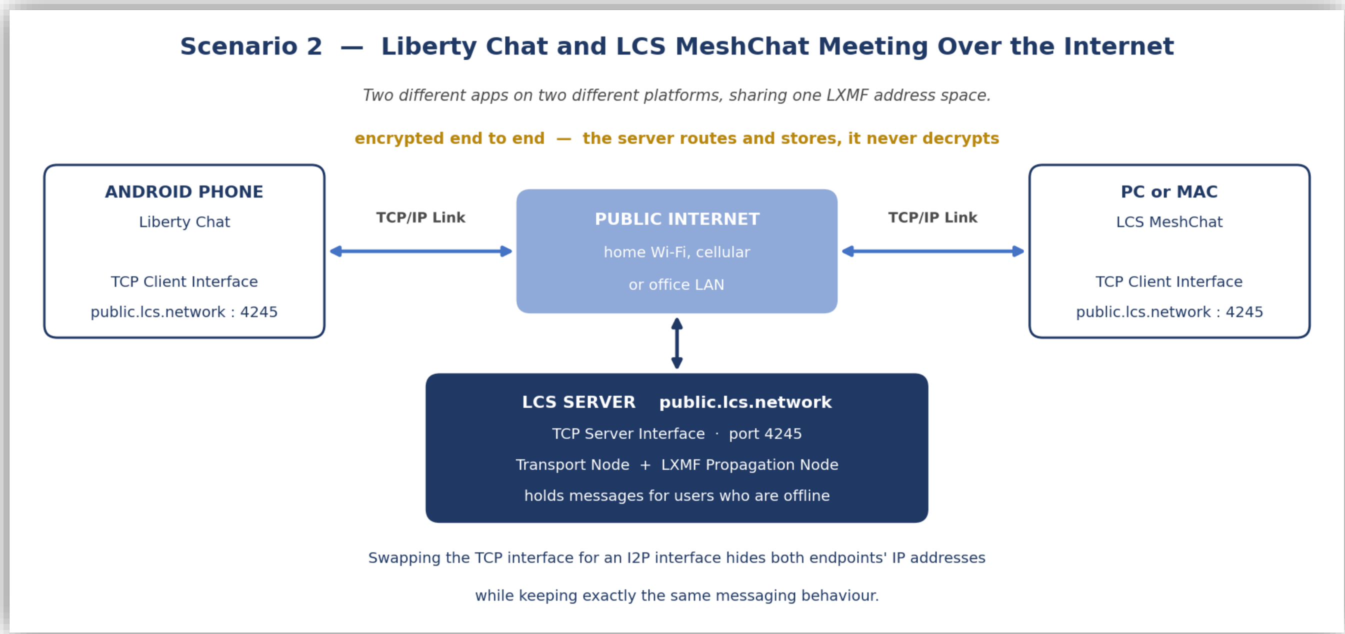

Figure 2 — Two different apps meeting on the LCS TCP server.

What you need

- An Android phone running Liberty Chat with any internet connection
- A Windows or Mac machine running LCS MeshChat
- A TCP Client interface on each, pointed at the server
- No radio hardware at all

How it works

- Both apps open an outbound TCP connection to the LCS server. Neither needs a public IP, a static address, or port forwarding.
- The server is a transport node, so it learns both announces and can route between them.
- It is also a propagation node, so a message sent while the other party is offline is stored and delivered on their return.
- The two apps are different programs on different operating systems, but both speak LXMF — so contacts, delivery receipts, attachments and voice all work between them.

Practical notes

- This is the easiest way to test the network before any hardware arrives.
- Bandwidth is plentiful here — use the **Opus high-bandwidth** voice option and full-quality live calls.
- For privacy, swap the TCP interface for an **I2P** interface. The apps behave identically, but neither endpoint's IP address is exposed.
- The server routes and stores. It never holds a key that can decrypt your messages.

Scenario 3 — Two mobile Command Center Gateways over packet radio

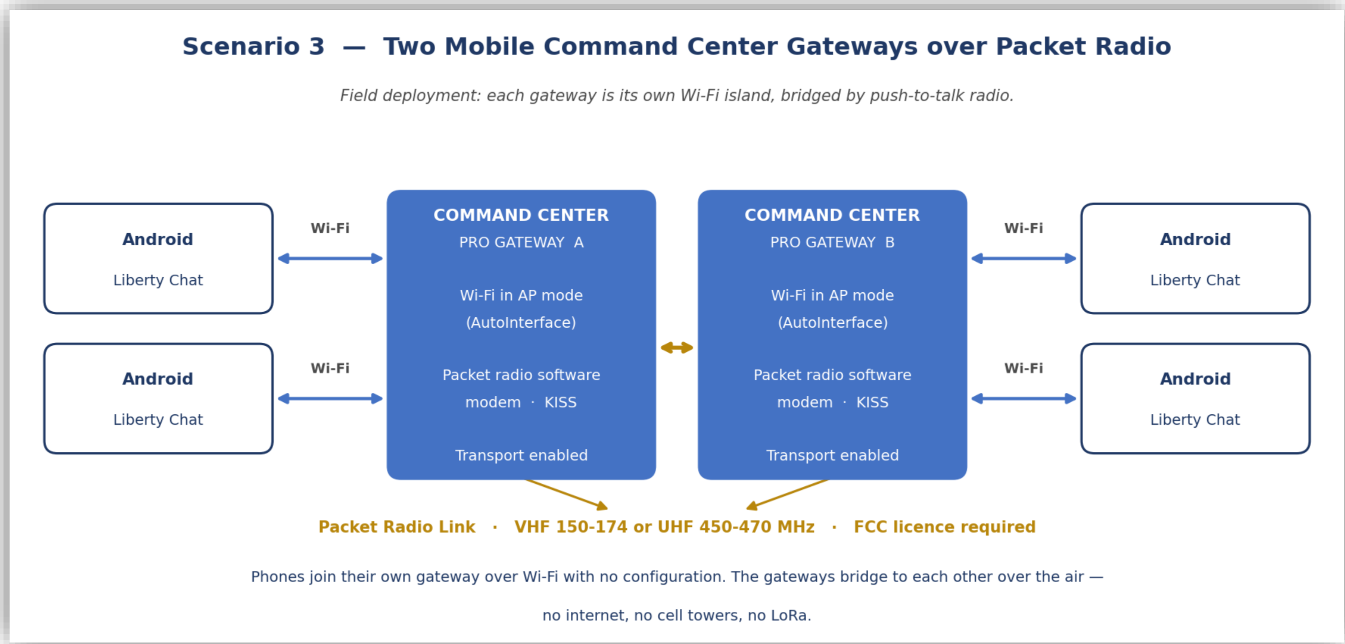

Figure 3 — Field deployment with no internet, no cell service and no LoRa.

What you need

- Two Command Center PRO Gateways, each in **Wi-Fi AP mode**
- A push-to-talk radio on each gateway, driven by the built-in software modem and PTT interface
- Androids running Liberty Chat at each site, joined to their local gateway's Wi-Fi
- An FCC license for encrypted operation on 150–174 MHz or 450–470 MHz

How it works

- Each gateway broadcasts its own Wi-Fi network. Phones join it and reach the gateway over **AutoInterface** with no setup — no IP configuration, no pairing.
- Both gateways have transport enabled, so each relays for the phones behind it.
- The gateways link to each other over the **packet radio** channel using KISS framing through the software modem.
- Announces from Site A cross the RF link and appear on the phones at Site B, and messages route automatically in both directions.

Practical notes

- Packet radio reaches much further than LoRa on comparable power and terrain, which is what makes this the long-haul option for a field deployment.
- It is also slower and half-duplex, so expect text and PTT voice clips rather than live calls across the RF hop.
- Each gateway holds mail for its own site. A phone that goes flat gets its messages when it rejoins the gateway Wi-Fi.
- Give the gateway interfaces a shared **network name and passphrase** if you want the deployment closed to outsiders on the same channel.

Scenario 4 — Two homes with IP RNodes on the local Wi-Fi

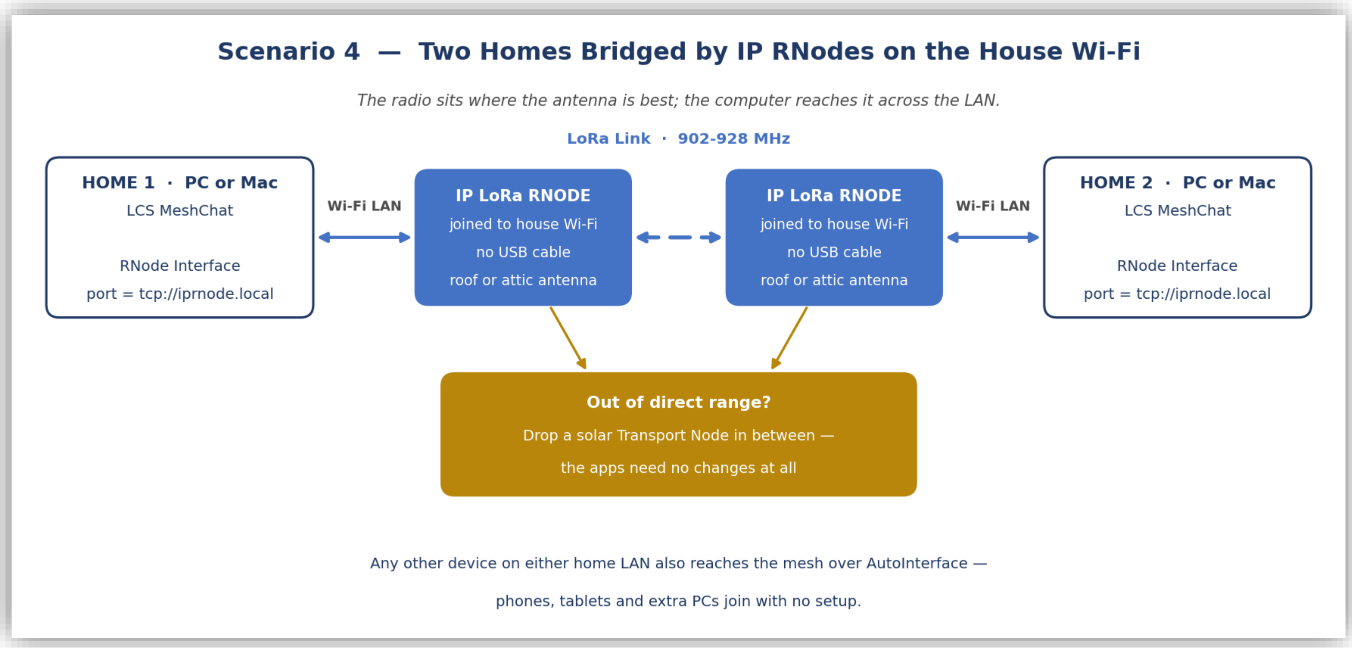

Figure 4 — The radio near the antenna, the app anywhere on the LAN.

What you need

- One IP LoRa RNode per house, joined to that house's Wi-Fi
- LCS MeshChat on a PC or Mac in each house
- An RNode interface in each MeshChat configured with **port = tcp://iprnode.local** (or the node's IP address)

How it works

- The RNode is reached over the LAN instead of a USB cable, so it can sit in the attic or by the window where the antenna performs best while the computer stays at your desk.
- MeshChat opens the radio across Wi-Fi and treats it exactly as if it were plugged in.
- The two RNodes talk to each other directly over LoRa, carrying the LXMF traffic between the houses.

Practical notes

- Any other device on either LAN — phones, tablets, a second PC — can also join over **AutoInterface** with no configuration, and will reach the mesh through whichever machine is connected to the radio.
- If the houses are out of direct LoRa range, add a solar transport node between them. Neither app changes; the path simply becomes two hops.
- Leave one machine running MeshChat as a **propagation node** so nobody misses messages while their computer is off.
- If either RNode is on a different subnet or the **.local** name does not resolve, use the node's IP address instead.

Scenario 5 — Two RNode user clusters joined by packet radio

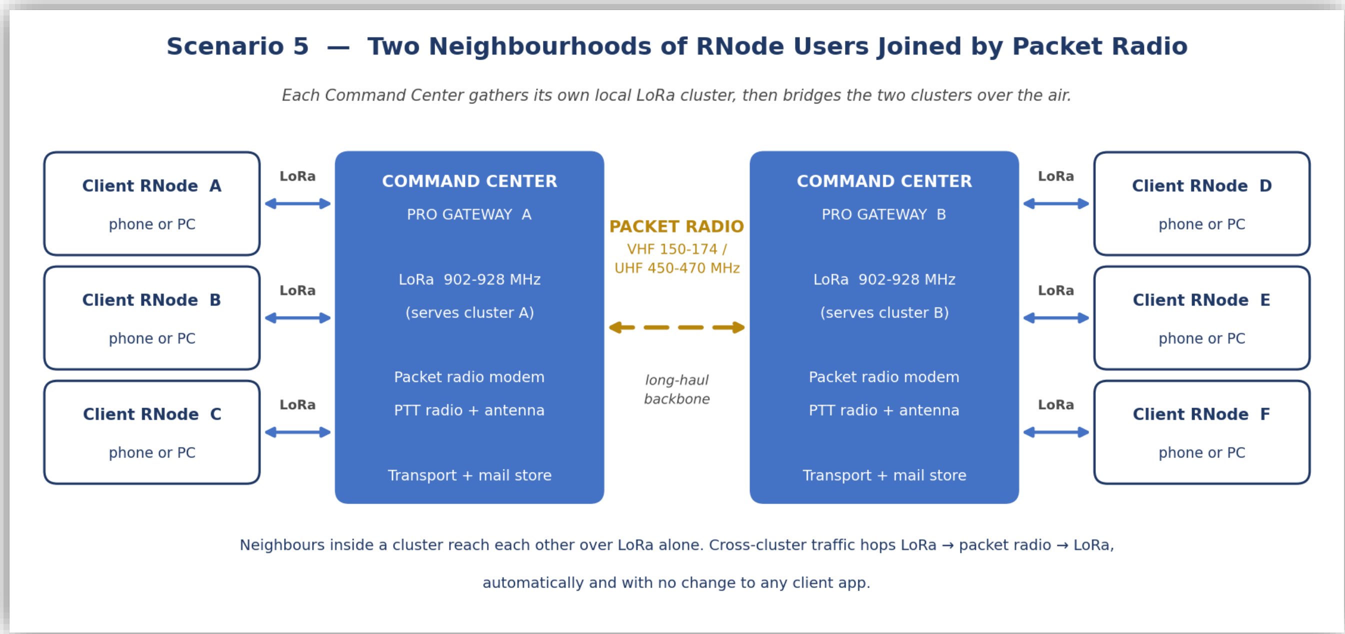

Figure 5 — Two local LoRa neighborhoods bridged into one network over the air.

This is the pattern for connecting two towns, two ends of a valley, or a base camp and a forward site. Each Command Center is the hub of its own LoRa neighborhood, and the packet radio channel carries traffic between the two hubs.

What you need

- Two Command Center PRO Gateways, sited for the best antenna position at each location
- A push-to-talk radio and antenna on each gateway, driven by the built-in software modem and PTT interface
- A cluster of LCS Client RNodes at each site — one per user, on USB-C or Bluetooth to a phone or PC
- Liberty Chat or LCS MeshChat on the user devices
- An FCC licence for encrypted operation on the business band

How it works

- **Inside a cluster,** users reach the gateway — and often each other — directly over **LoRa**. Local chat never touches the packet radio channel at all.
- **Between clusters,** the two gateways bridge LoRa to packet radio. A message from user A to user D crosses three hops: LoRa in, packet radio across, LoRa out.
- Announces propagate the same way, so users at Site A simply see the Site B users appear in their peer list by name. Nobody selects a route or a path.
- Both gateways run as **transport nodes** and as **propagation nodes**, so each site also holds mail for its own users.

Why this topology

- **It plays to each radio's strength.** LoRa is license-free and ideal for short, dense, local links. Packet radio reaches far further on comparable power, which makes it the right choice for the single long hop between sites.
- **It keeps the slow link quiet.** Most traffic is local and stays on LoRa, so the RF backbone only carries what actually has to cross.
- **It scales by adding gateways.** A third neighborhood joins by putting a third Command Center on the same packet radio channel — no reconfiguration at any client.

- **It survives loss.** If the packet radio link drops, each cluster keeps working normally and cross-site messages queue at the propagation node until the link returns.

Practical notes

- Keep every RNode in a cluster on the same frequency, bandwidth, spreading factor and coding rate. A mismatch produces silence, not a slow link.
- Expect text and push-to-talk voice clips across the RF backbone. Live LXST calls between sites will be marginal — inside a cluster on a fast LoRa preset they are more realistic.
- If range within a cluster is short, add a solar transport node to extend the LoRa side. The gateway and the packet radio link are unaffected.
- Set a shared **network name and passphrase** on the gateway interfaces to close the deployment to anyone else on the same channel.

16. What Your Network Requires — Checklist

Every device, without exception

- An **identity** — generated on first launch, backed up somewhere safe
- At least one **interface** — a radio, a Wi-Fi connection, or a TCP server address
- An **announce** — press it once at setup, and again whenever you change how you connect

For a group that is not always online

- At least one **propagation node** that stays powered — a Command Center Gateway, the LCS server, or a PC running MeshChat

For range beyond a single radio hop

- One or more **transport nodes** with **enable_transport = Yes**, sited high
- Matching radio parameters on every unit on that channel — frequency, bandwidth, spreading factor, coding rate. A mismatch means silence, not a slow link.

For reaching the wider world

- A **TCP or I2P interface** on at least one device that has internet access. Everything behind it comes along.

For live voice

- A fast enough link: Wi-Fi, Ethernet, TCP, or LoRa at **Short Fast** or better. Otherwise use push-to-talk voice clips.

For legal operation

- **LoRa at 902–928 MHz** — license-free in the United States
- **Packet radio on 150–174 or 450–470 MHz** — FCC license required for encrypted business-band use
- Rules differ by country. Confirm what applies where the equipment is operated.

What you do not need

- No accounts, no subscriptions, no SIM cards, no phone numbers
- No DNS, DHCP, port forwarding or static IP addresses
- No permission from anyone to join, extend, or leave the network

Reference Links

- Reticulum Network Stack manual — https://github.com/markqvist/Reticulum
- LXMF messaging protocol — https://github.com/markqvist/LXMF
- LXST signal transport protocol — https://github.com/markqvist/LXST
- Liberty Chat (Columba distribution) — https://github.com/daylight-hub/columba
- LCS MeshChat — https://github.com/daylight-hub/reticulum-meshchat
- Liberty Communication Systems, Inc. — https://lcs.network/